

ME 206P: Mechanics of Solid Lab

Course Code: ME 206P

Course Name: Mechanics of Solid Lab

L-T-P-C: 0-0-2-1

Prerequisites: ME 206 Mechanics of Solids

Intended for: B.Tech.

Mutual Exclusion: None

Approval: 61st BoA (replaces ME 311P Design Lab-I)

Preamble

The proposed course aims to provide students with a strong foundation in the experimental aspects of mechanics of solids by linking theoretical principles with practical testing and analysis. Through a series of well-structured laboratory experiments, students will learn to evaluate material properties, analyze deformation and failure modes, and interpret experimental data with engineering relevance. This hands-on experience will develop experimental skills, improve understanding of stress-strain relationships, and strengthen problem-solving capabilities. The course will also expose students to standard testing equipment and procedures, fostering familiarity with industrial practices. Overall, the course bridges classroom learning with real-world engineering applications, enhancing both conceptual clarity and practical competence.

Course Modules

1. **Torsion Test – Modulus & Shear Stress:** Conduct torsion test on cylindrical specimens to determine modulus of rigidity and shear stress-strain relationships. The experiment develops understanding of torsional deformation, material strength, and failure modes in shafts, directly applicable to real-life machine and structural components.
2. **Rockwell Hardness Determination:** Perform Rockwell hardness test on metallic samples to evaluate indentation hardness. This quick, nondestructive test helps in material classification, quality control, and correlation with mechanical properties such as tensile strength and wear resistance.
3. **Bending Test – Young’s Modulus:** Determine Young’s modulus of a specimen using a three-point bending setup. The test also allows measurement of force-deflection characteristics of beams, enhancing understanding of flexural stiffness, elastic recovery, and deformation behavior under transverse loading.
4. **Photoelastic Stress Concentration Analysis:** Use photoelasticity techniques to visualize stress distribution around holes or notches. This experiment highlights stress concentration phenomena and validates theoretical predictions, enabling students to appreciate optical methods for stress analysis in complex geometries.
5. **Spring Stiffness & Strain Energy:** Measure stiffness of a helical spring and modulus of rigidity of the spring wire. The experiment further determines strain energy storage capacity, linking to vibration absorbers and energy storage devices in engineering applications.
6. **Friction & Wear Characterization:** Study frictional behavior of material pairs under varied loads, speeds, and lubrication conditions. Additionally, measure wear characteristics to understand tribological performance. This experiment provides insights into machine component life, efficiency, and surface engineering strategies.

7. **Izod/Charpy Impact Testing:** Evaluate toughness, hardness, and impact resistance of materials using standard Izod/Charpy tests. The experiment demonstrates material response to sudden loading and highlights differences between brittle and ductile fracture behavior.
8. **Thin Cylinder Stress–Strain Study:** Perform stress–strain analysis on thin-walled metallic cylinders under internal pressure. The test illustrates circumferential and longitudinal stresses, validating thin-shell theory and its applications in pressure vessels and pipelines.
9. **Column Buckling Behaviour Analysis:** Investigate buckling of columns with different end conditions. Predict critical load and evaluate Young’s modulus from buckling data. This experiment emphasizes stability, slenderness ratio effects, and structural design considerations.
10. **Surface Roughness Measurement:** Use a surface roughness tester to evaluate surface finish of specimens. This test highlights the role of machining parameters, surface texture, and their influence on fatigue strength, wear, and lubrication performance.
11. **Tensile Test – Nonferrous Specimen:** Conduct tensile test on nonferrous specimens and compare stress–strain curves with ferrous materials. This experiment highlights differences in ductility, yield strength, and work-hardening behavior, providing comparative understanding of material classes.

Text Books

1. Timoshenko S. P., and Gere J. M., *Mechanics of Materials*, 2nd Edition, CBS Publishers, 2002.
2. Crandall S. H., Dahl N. C., and Lardner T. J., *An Introduction to the Mechanics of Solids*, 2nd Edition., McGraw-Hill, 1999.

References

1. Hearn E. J., *Mechanics of Materials*, 3rd Edition, Pergamon, 2003.
2. Higdon A., Ohlsen E. H., Stiles W. B., Weese J. A., and Riley W. F., *Mechanics of Materials*, John Wiley & Sons, 1989.
3. Popov E. P., Nagarajan S., and Lu Z. A., *Mechanics of Materials*, 2nd Edition, Prentice-Hall of India, 2002.